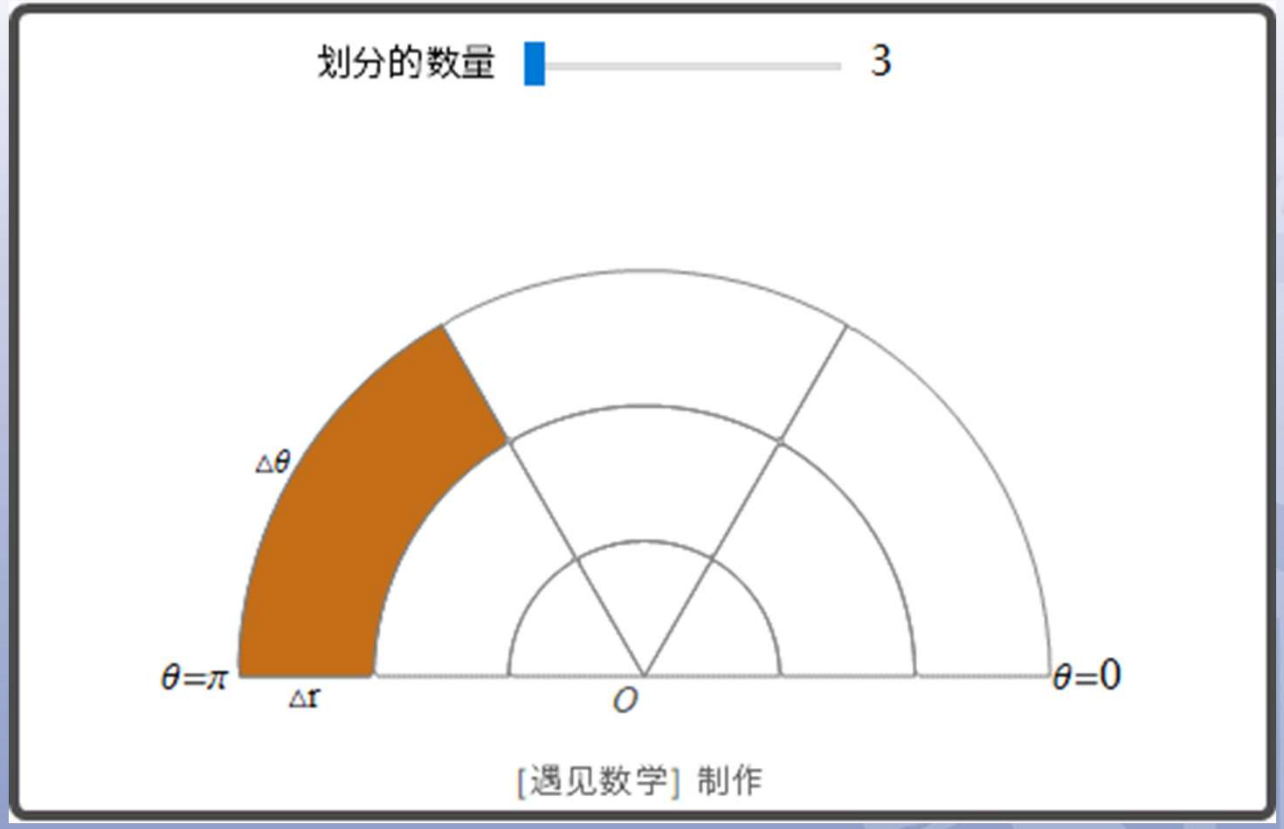
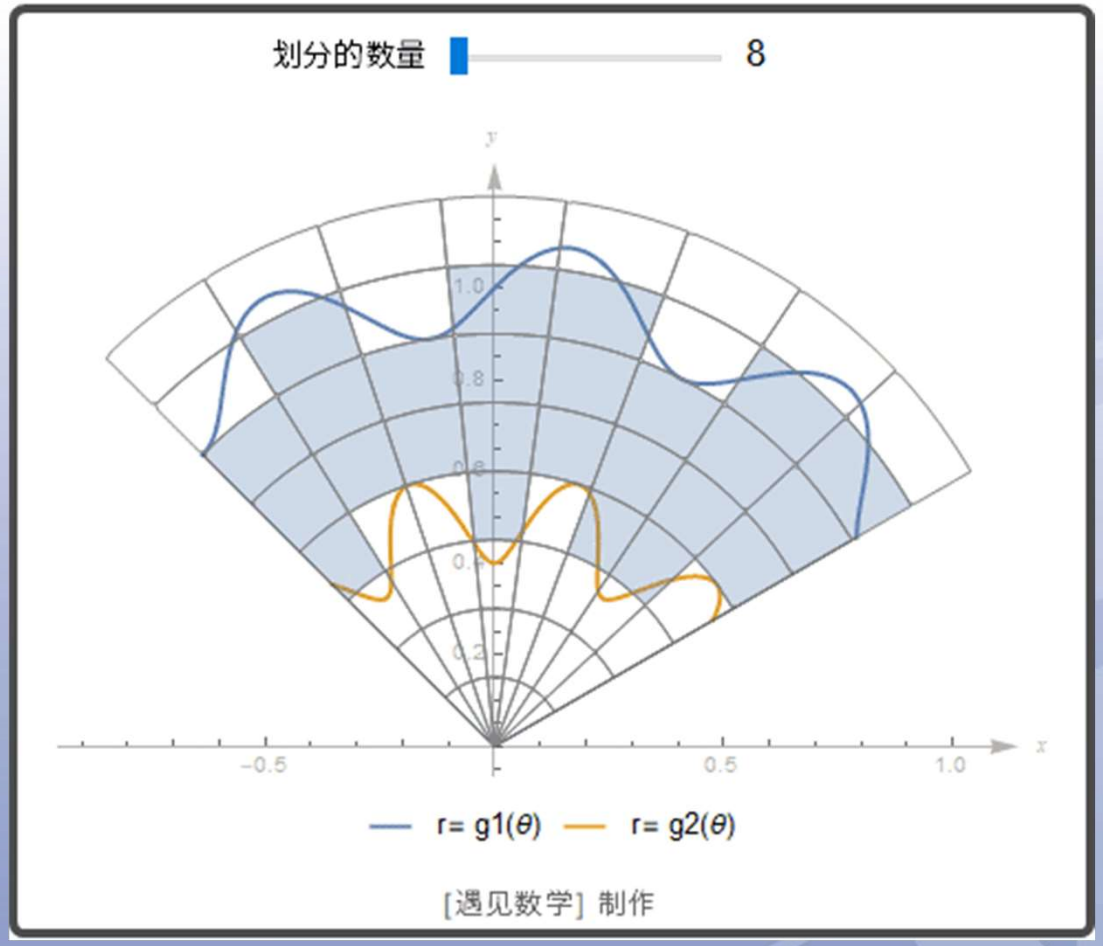
二、利用极坐标系计算二重积分



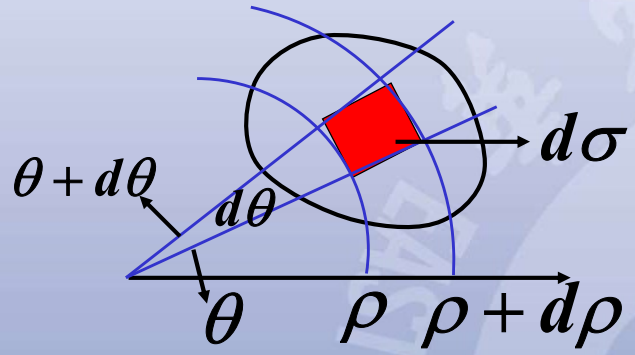

划分的数量
8
y
1.0
0.8
0.6
0.4
0.2
-0.5
0.5
1.0
x
r= g1(θ)
r= g2(θ)
[遇见数学] 制作

二、利用极坐标系计算二重积分

在极坐标系中,设D的边界与过极点的射线相交不多于两点,

用过极点的射线和以极点为圆心的圆周将D分成若干子域,
如图可知:

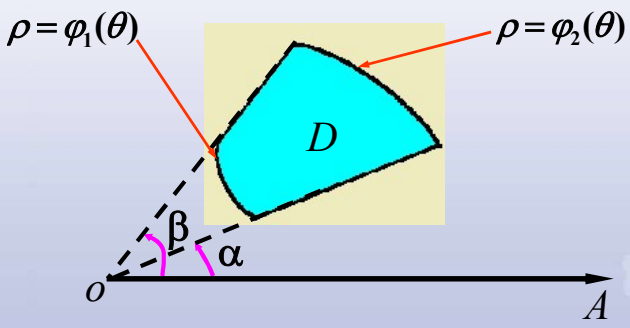
$$d\sigma = \rho d\theta d\rho$$



$$\iint_D f(x,y)d\sigma = \iint_D f(\rho\cos\theta, \rho\sin\theta)\rho d\rho d\theta$$

二重积分化为二次积分的公式（1）

区域特征如图



$$\alpha \le \theta \le \beta,$$

$$\varphi_1(\theta) \le \rho \le \varphi_2(\theta).$$

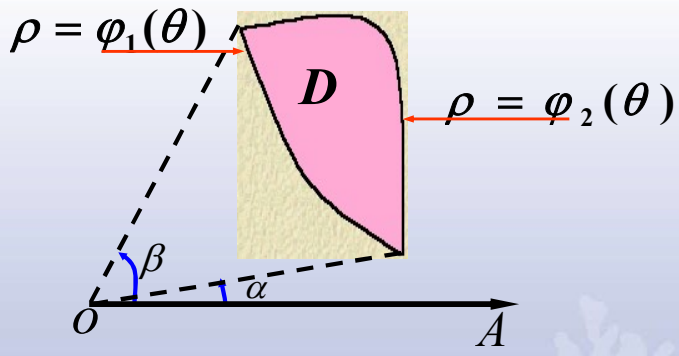
$$\iint_D f(\rho\cos\theta, \rho\sin\theta)\rho d\rho d\theta$$

$$= \int_\alpha^\beta d\theta \int_{\varphi_1(\theta)}^{\varphi_2(\theta)} f(\rho\cos\theta, \rho\sin\theta)\rho d\rho.$$

区域特征如图

$$\alpha \le \theta \le \beta,$$

$$\varphi_1(\theta) \le \rho \le \varphi_2(\theta).$$



$$\iint_D f(\rho\cos\theta, \rho\sin\theta)\rho d\rho d\theta$$

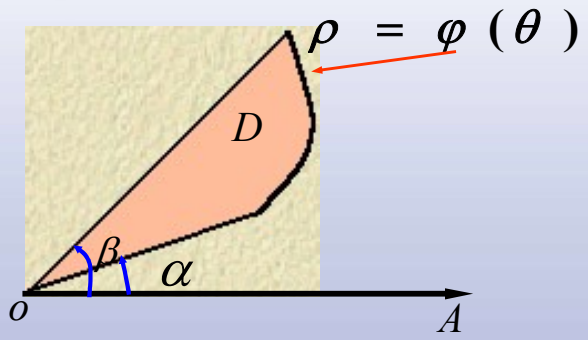
$$= \int_\alpha^\beta d\theta \int_{\varphi_1(\theta)}^{\varphi_2(\theta)} f(\rho\cos\theta, \rho\sin\theta)\rho d\rho.$$

二重积分化为二次积分的公式（２）

区域特征如图

$$\alpha \le \theta \le \beta,$$

$$0 \le \rho \le \varphi(\theta).$$

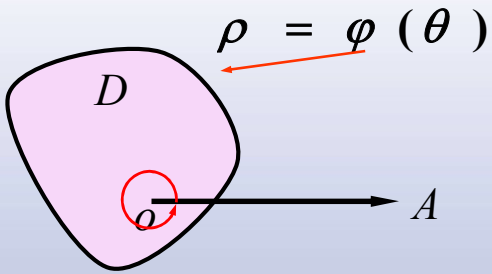


$$\iint_D f(\rho\cos\theta, \rho\sin\theta)\rho d\rho d\theta$$

$$= \int_\alpha^\beta d\theta \int_0^{\varphi(\theta)} f(\rho\cos\theta, \rho\sin\theta)\rho d\rho.$$

二重积分化为二次积分的公式（3）

区域特征如图



$$0 \le \theta \le 2\pi, \quad 0 \le \rho \le \varphi(\theta).$$

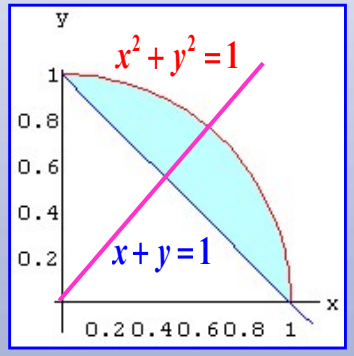
$$\iint_D f(\rho\cos\theta, \rho\sin\theta)\rho d\rho d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^{\varphi(\theta)} f(\rho\cos\theta, \rho\sin\theta)\rho d\rho.$$

例 1 写出积分$\iint\limits_D f(x,y)dxdy$的极坐标二次积分形式，其中积分区域

$$D=\{(x,y)\mid 1-x\le y\le\sqrt{1-x^2},\ 0\le x\le 1\}.$$

解 在极坐标系下$\begin{cases} x=\rho\cos\theta \\ y=\rho\sin\theta \end{cases}$



所以圆方程为 $\rho=1$,

直线方程为$\rho=\dfrac{1}{\sin\theta+\cos\theta}$,

$$\iint\limits_D f(x,y)dxdy=\int_0^{\frac{\pi}{2}}d\theta\int_{\frac{1}{\sin\theta+\cos\theta}}^{1}f(\rho\cos\theta,\rho\sin\theta)\rho d\rho.$$

说明： 计算二重积分时，

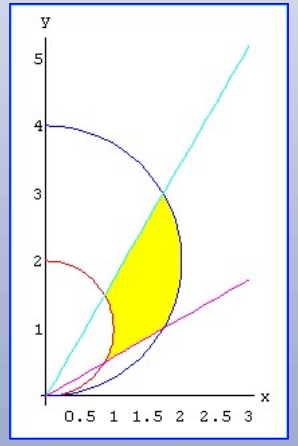
若被 积函数中含有 x^2+y^2 或 $\frac{y}{x}$,

积分区域的边界曲线方 程中含有 x^2+y^2 时，

常使用极坐标变换.

例 7　计算$\iint\limits_D (x^2+y^2)dxdy$，其 D 为由圆 $x^2+y^2=2y$，$x^2+y^2=4y$及直线$x-\sqrt{3}y=0$，$y-\sqrt{3}x=0$ 所围成的平面闭区域.

解



$$y-\sqrt{3}x=0 \Rightarrow \theta_2=\frac{\pi}{3}$$

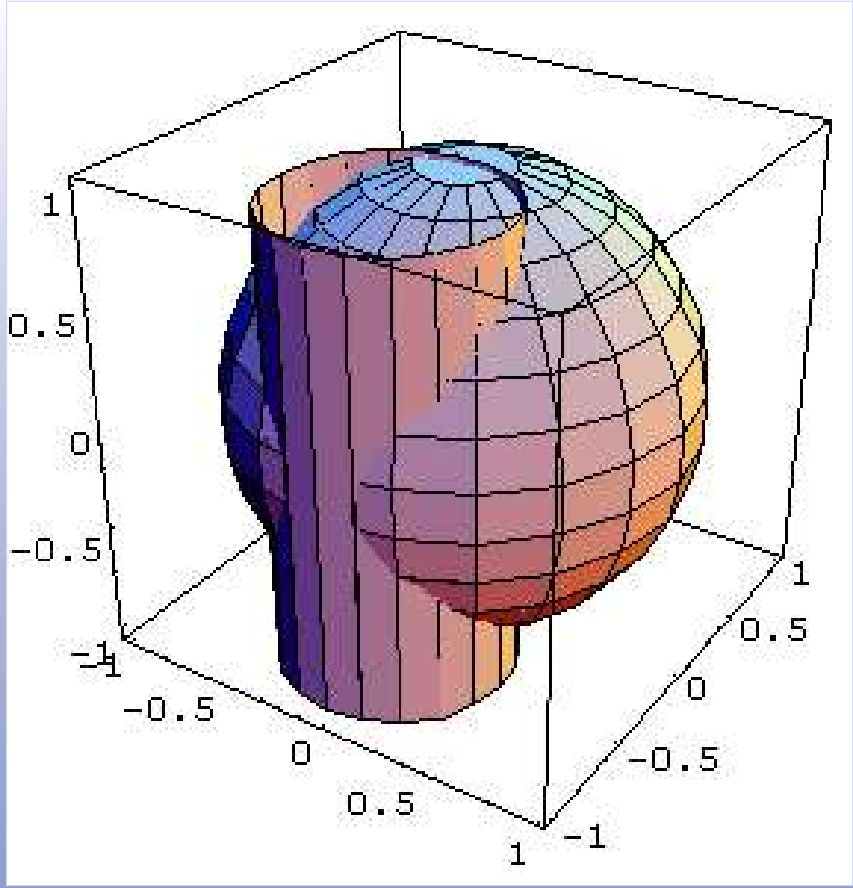
$$x^2+y^2=4y \Rightarrow \rho=4\sin\theta$$

$$x-\sqrt{3}y=0 \Rightarrow \theta_1=\frac{\pi}{6}$$

$$x^2+y^2=2y \Rightarrow \rho=2\sin\theta$$

$$\iint\limits_D (x^2+y^2)dxdy=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}d\theta\int_{2\sin\theta}^{4\sin\theta}\rho^2\cdot\rho d\rho=15(\frac{\pi}{2}-\sqrt{3}).$$

$x^2+y^2+z^2=4a^2 \quad x^2+y^2\le 2ax$ 围成图形的体积

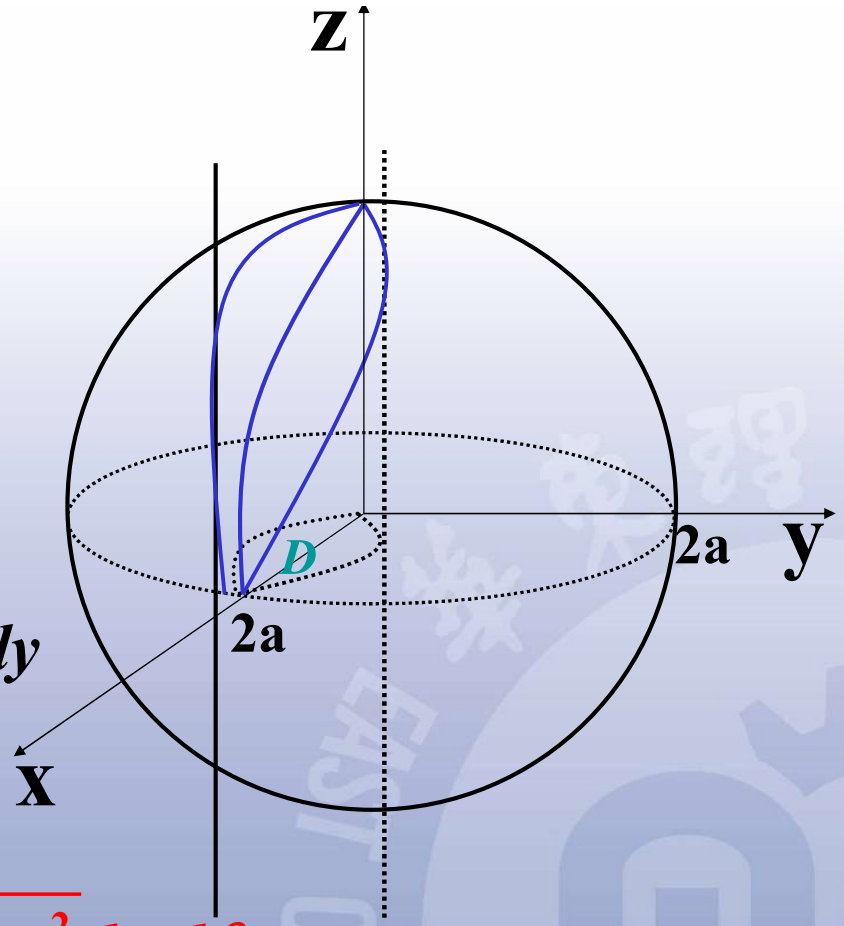


$$z=\sqrt{4a^2-x^2-y^2}$$

$$D: x^2+y^2\le 2ax$$

$$V=2\iint_D z\,dxdy$$

$$=4\iint_{D_1}\sqrt{4a^2-x^2-y^2}\,dxdy$$



$$=4\iint_{\left\{\begin{array}{c}0\le\theta\le\frac{\pi}{2}\\ 0\le\rho\le 2a\cos\theta\end{array}\right\}}\rho\sqrt{4a^2-\rho^2}\,d\rho\,d\theta$$

例　计算 $\iint\limits_{x^2+y^2\le R^2}\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)dxdy$

解 $\iint\limits_{x^2+y^2\le R^2} x^2dxdy \xlongequal{\text{交换 } x,y} \iint\limits_{y^2+x^2\le R^2} y^2dxdy$

$\xlongequal{\text{积分区域不变}} \iint\limits_{x^2+y^2\le R^2} y^2dxdy$

$$\text{原式}=\frac{1}{a^2}\iint\limits_{x^2+y^2\le R^2} x^2dxdy+\frac{1}{b^2}\iint\limits_{x^2+y^2\le R^2} y^2dxdy$$

$$=\left(\frac{1}{a^2}+\frac{1}{b^2}\right)\iint\limits_{x^2+y^2\le R^2} x^2dxdy$$

$$=\left(\frac{1}{a^2}+\frac{1}{b^2}\right)\iint\limits_{x^2+y^2\le R^2} x^2\,dxdy$$

$$=\frac{1}{2}\left(\frac{1}{a^2}+\frac{1}{b^2}\right)\iint\limits_{x^2+y^2\le R^2} (x^2+y^2)\,dxdy$$

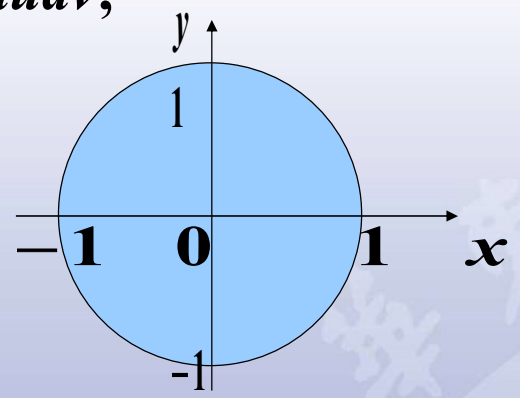
$$=\frac{1}{2}\left(\frac{1}{a^2}+\frac{1}{b^2}\right)\int_0^{2\pi} d\theta\int_0^{R}\rho^3 d\rho$$

$$=\frac{\pi}{4}R^4\left(\frac{1}{a^2}+\frac{1}{b^2}\right)$$

若连续函数$f(x,y)$满足

$$f(x,y)=5(x^2+y^2)^{\frac{3}{2}}-\iint_{u^2+v^2\le 1} f(u,v)dudv,$$

则 $\iint_{x^2+y^2\le 1} f(x,y)dxdy=$ $\underline{\dfrac{2\pi}{\pi+1}}$



解：积分区域 D 如图所示，

等式两边同时在 D 上求积分，

记所求积分 $\iint_{x^2+y^2\le 1} f(x,y)dxdy=I$，则

$$I=5\iint_{x^2+y^2\le 1}(x^2+y^2)^{\frac{3}{2}}dxdy-\iint_{x^2+y^2\le 1} Idxdy$$

$$(\pi+1)I=5\int_0^{2\pi}d\theta\int_0^1\rho^3\cdot\rho\,d\rho\ =2\pi$$

$$\therefore I=\frac{2\pi}{\pi+1}$$